

Cell
- Content: Object - isWalkable: boolean
• setContent(content: Object): void • clearContent(): void • isBlocked(): boolean

Moving Enemy
- positionX: int - positionY: int - Target: Doug
• move(): void • Chase(Doug: Doug): void

Punishment
- penaltyPoints: int - Type: String
• applyPenalty(Doug: Doug): void

GameMechanics
- Score: int - TimeElapsed: int - gameState:String
• startGame(): void • endGame(): void • checkVictory(): boolean • checkLoss(): boolean

Doug
- positionX: int - positionY: int - score: int - collectedFood: List<Reward> - health: int
• move(direction: String): void • collectReward(reward: Reward):void • Penalize(punishment: Punishment): void

Enemy
- positionX: int - positionY: int - Type: String - Image: String
• Move(): void • Interact(Doug: Doug): void

Item
- Points: int - Image: String
• Collect(): int

Reward
- Points: int - Type: String - Image: String
• Collect(): int

BonusReward
- bonusPoints: int
• collectBonus(): int

Cat
- Move(): void - stealFood(Doug: Doug): void

Rat
- Move(): void - stealFood(Doug: Doug): void

Obstacle
- Type:String
• blackMovement(): boolean

Map
- Cells: 2D array<Cell> - mapName: String - startPoint: Cell - endPoint: Cell
• loadMap(level: int): void • renderMap(): void • updateCell(x: int, y: int, content: Object): void

